Climate Change Analysis and Temperature Prediction Using Machine Learning

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Introduction

Climate change is one of the most urgent global challenges because rising temperatures affect agriculture, water resources, biodiversity, public health, infrastructure resilience, and long-term environmental planning. Temperature prediction is therefore not only a scientific forecasting task but also a practical requirement for planners, governments, and communities that must respond to climatic change through adaptation and risk management. Recent research shows that machine learning models can learn nonlinear climatic relationships from large environmental datasets and often outperform simple statistical baselines in complex prediction settings [1], [3]. For this reason, temperature modeling has become an important research area at the intersection of climate science, machine learning, and applied data analytics.

The problem, however, is not merely to predict temperature values; it is to predict them reliably across different regions, time scales, and data conditions while preserving interpretability. This study uses the Berkeley Earth Climate Change dataset, comprising approximately 8.6 million city-level monthly temperature records spanning from 1750 to 2013, as the primary data source for analysis and model development. The scale and historical depth of this dataset make it especially useful for studying long-term warming behavior, seasonal patterns, and geographically distributed temperature variation. Existing studies examine seasonal anomalies, high-frequency local climate signals, urban thermal environments, semi-arid forecasting, and regional climate time series [2], [4], [5], [7]. Although these studies report encouraging accuracy, they differ substantially in dataset scope, feature engineering, validation design, and performance metrics.

A more specific gap emerges from this variation. Many published studies are based on specialized sensors, remote-sensing pipelines, narrow regional case studies, or short operational horizons, whereas relatively few compare multiple machine learning models on a large historical city-level climate dataset using one transparent workflow [1], [5], [8]. Several studies also emphasize prediction accuracy without clearly linking model outputs to interpretable climate-change signals such as anomaly behavior, regional warming patterns, or temporally ordered validation. In climate-related modeling, this is a serious limitation, because a model can appear accurate under convenient evaluation conditions yet still provide weak evidence for realistic long-term generalization.

This study is significant because it addresses methodological and analytical weaknesses in the literature while aligning directly with a complete data science workflow. By using a real historical city-level dataset, engineering climate-relevant features, comparing multiple machine learning models, and emphasizing anomaly analysis and chronological validation, the research aims to produce findings that are both technically reliable and meaningful for climate-change interpretation. The expected contribution is not only the identification of a strong predictive model, but also a clearer understanding of how feature engineering, validation strategy, and regional analysis affect temperature prediction performance. It therefore contributes a more reproducible, comparative, and regionally relevant framework for machine-learning-based temperature prediction.

Related Work

3.1 Large-Scale Climate Forecasting Studies

A major strand of the literature examines temperature prediction at regional or long-horizon scales, where the central challenge is to capture nonlinear climatic behavior over extended time periods. Mishra et al. compared SARIMA, TDNN, LSTM, and XGBoost on centennial rainfall and temperature series across South Asia and showed that no single model consistently dominates; rather, performance varies by variable type, country context, and the degree of nonlinearity in the data [1]. Ratnam et al. addressed seasonal air-temperature anomalies by applying a genetic algorithm to optimize the weights of ensemble climate-model members, demonstrating that machine learning can improve forecast skill even when it is used to enhance physical climate-model outputs rather than replace them [4]. These studies agree that adaptive and data-driven methods are increasingly necessary for climatic forecasting, but they differ in whether predictive gains are achieved through standalone machine learning models or through optimization of existing numerical systems.

3.2 Local and High-Resolution Prediction Studies

A second cluster of research focuses on local, high-resolution, or operationally specialized prediction contexts. Zanchi et al. demonstrated that deep learning can model local microclimate effectively and found that local meteorological inputs yield more accurate forecasts than coarse global data, although reanalysis products remain viable when local observations are unavailable

[2]. Khan and Al-Hajj used five-minute climate observations in Kuwait and reported that FTTransformer and LSTM outperformed traditional machine learning models in simultaneous prediction of multiple air and surface temperature targets [3]. Adeniran et al. similarly achieved strong results through a federated learning neural network for spatially continuous near-surface air-temperature estimation and urban heat-island analysis [5]. These studies share a common trend: richer and more granular inputs often improve performance. At the same time, their strongest results frequently depend on dense sensor networks, remote-sensing products, or geographically limited datasets, which restricts direct transfer to long-term historical city-level climate analysis.

3.3 Machine Learning Model Comparison Studies

The comparative machine learning literature is especially useful for identifying broader patterns in model behavior. Abdelsattar et al. benchmarked nine machine learning models for temperature and humidity prediction and concluded that XGBoost and other ensemble-based approaches substantially outperformed linear and support vector methods, reinforcing the practical strength of boosting in environmental prediction tasks [6]. Elbeltagi et al. compared five data-driven models for daily minimum and maximum temperature estimation in Egypt and found that M5P outperformed linear regression, additive regression, random subspace, and SVM under their selected feature combinations [7]. The broad agreement across these studies is that simple linear models remain valuable as interpretable baselines but generally provide weaker predictive performance than nonlinear and ensemble-based methods. However, there is no consistent agreement on a single best model, because model superiority changes with dataset structure, feature design, forecasting horizon, and regional climate conditions. This inconsistency strengthens the need for a controlled comparative study on a single large historical dataset.

3.4 Evaluation Strategies, Implementation, and Limitations

Another important trend concerns evaluation methodology and the limited translation of research into accessible real-world systems. Most of the reviewed studies use MAE, RMSE, MSE, correlation, or R2-style indicators, but their validation strategies differ considerably. Some rely on standard train-test partitions [6], [7], some use cross-validation [1], [5], and others adopt temporally stronger procedures such as leave-one-year-out validation or seasonal ensemble optimization [3], [4]. This distinction matters because a model that performs well under random partitioning may not generalize equally well under temporally ordered testing, particularly in climate applications where temporal leakage can distort conclusions. An ACM study by Nirere et al. is useful here because it moves beyond pure prediction and presents an IoT-based climate change prediction system that combines sensors, MQTT communication, MongoDB storage, and machine learning for climate-warming assessment in Rwanda [8]. Although its focus is broader than temperature forecasting and its methodology is more system-oriented than comparative, it highlights an important trend: real-world climate solutions often emphasize deployment and monitoring, while comparative temperature-model evaluation remains underdeveloped. Taken together, the literature confirms the value of machine learning for temperature prediction, but it also reveals the need for a more accessible, comparative, and interpretable framework that combines historical city-level data, anomaly-based interpretation, realistic validation, and clearer research-to-application linkage.

Research Gap

The reviewed literature shows that machine learning can improve temperature prediction, yet existing studies remain fragmented across specialized datasets, narrow regional case studies, and inconsistent validation procedures. There is still limited evidence from a single reproducible workflow that compares multiple machine learning models on long-term historical city-level temperature data while also incorporating anomaly-based interpretation, temporally realistic validation, and a clearer bridge to practical climate-analysis use. This unresolved issue forms the central research gap addressed by the present study.

Research Question

This study addresses the following central research question:

Which machine learning model most effectively predicts city-level temperature from historical climate data, and how does predictive performance compare across models when evaluated using MAE, RMSE, and R2 under a common experimental framework?

Sub-questions:

1. How do engineered temporal and geographic features affect the accuracy of city-level temperature prediction?

2. How does chronological validation differ from random validation in estimating model generalization performance?

3. What temperature anomaly patterns are visible in South Asia, and how do model results relate to those regional trends?

Purpose

Novelty of the Study

The novelty of this study lies in combining city-level temperature prediction with anomaly-based climate interpretation, chronological validation, and a South Asia case study. This makes the work more than a basic model-comparison exercise because it connects predictive performance with temporal realism and regional climate relevance.

The purpose of this research is to develop a comparative and interpretable machine learning framework for analyzing climate change patterns and predicting city-level temperature from historical climate records. It aims to identify robust predictive models under a consistent methodology while generating findings that are meaningful for anomaly analysis, chronological validation, and regional climate interpretation.

Objectives

1. To analyze a large historical city-level climate dataset by examining its structure, temporal coverage, and quality before model development.

2. To preprocess the dataset by handling missing values, correcting date and coordinate formats, and standardizing relevant modeling inputs.

3. To engineer climate-relevant temporal and geographic features, including lag values, rolling averages, seasonal indicators, and temperature anomalies.

4. To perform exploratory data analysis including global temperature trend visualization, seasonal pattern analysis, correlation heatmap, and regional anomaly inspection to extract meaningful climate insights before modeling.

5. To develop and compare at least four machine learning regression models for monthly temperature prediction using a common experimental setting.

6. To optimize selected models through hyperparameter tuning and evaluate them using MAE, MSE, RMSE, MAPE, and R2.

7. To interpret model performance in relation to chronological validation, South Asia focused analysis, and broader climate-change trends.

Expected Outcomes

This research is expected to produce the following outcomes:

1. A fully cleaned and feature-engineered dataset ready for reproducible machine learning experimentation.

2. A comparative performance table showing MAE, MSE, RMSE, MAPE, and R2 scores for all implemented models.

3. Identification of the best-performing machine learning model for long-term historical city-level temperature prediction.

4. Visual outputs including temperature trend plots, correlation heatmaps, actual vs predicted graphs, residual error plots, and feature importance charts.

5. A South Asia regional analysis showing temperature anomaly trends and model behavior on a geographically specific subset.

6. Evidence-based conclusions linking model performance to climate-change interpretation and chronological validation results.

Methodology

The study will follow a structured data science methodology so that the research process remains transparent, reproducible, and aligned with the identified research gap as well as the actual project implementation.

Data Collection

The research will use the Berkeley Earth Climate Change dataset available on Kaggle, specifically the file GlobalLandTemperaturesByCity.csv, which contains approximately 8.6 million city-level monthly temperature records spanning 1750 to 2013 [9].

Data Preprocessing

The raw dataset will be cleaned through missing-value handling, duplicate removal, datetime parsing, and conversion of latitude and longitude into usable numeric form. A processed modeling subset will then be created from the cleaned records.

Feature Engineering

Relevant predictive variables will be derived from the cleaned data, including year, month, quarter, season, hemisphere, decade, lagged temperatures, rolling means, historical city-month averages, temperature anomaly, and a simple climate-risk-oriented index. A South Asia subset will also be created for regional analysis.

Model Development

The project will implement Linear Regression, Decision Tree Regressor, Random Forest Regressor, Gradient Boosting Regressor, and XGBoost Regressor. Both random and chronological split strategies will be used so that baseline performance can be compared against time-aware validation.

Evaluation

Selected models will be tuned through RandomizedSearchCV, and final performance will be compared using MAE, MSE, RMSE, MAPE, and R2. The analysis will also include cross-validation summaries, before-versus-after tuning comparison, feature-importance analysis, and decade-wise error examination so that the results remain both quantitative and interpretable. Parameters such as n_estimators, max_depth, learning_rate, and min_samples_split will be optimized for Random Forest and Gradient Boosting Regressor during tuning.

Reference List

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